

ANNAMALAI



UNIVERSITY

FACULTY OF SCIENCE

DEPARTMENT OF COMPUTER AND INFORMATION SCIENCE

PROJECT VIVA VOCE PRESENTATION

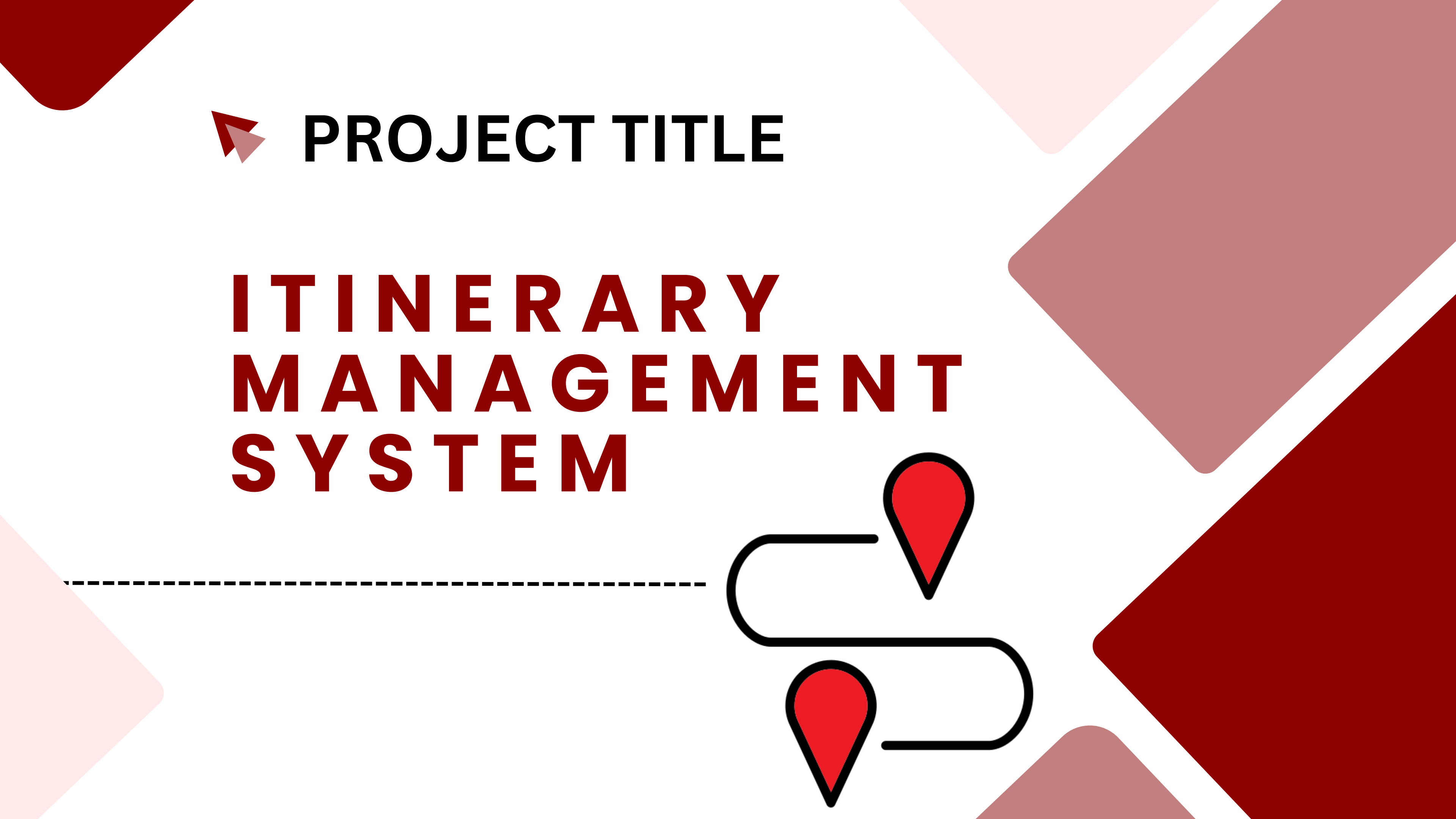
ITINERARY MANAGEMENT SYSTEM

PRESENTED BY :

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PROGRAMME : M.Sc. DATA SCIENCE

UNDER THE GUIDANCE OF :

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Assistant Professor/Programmer



➤ **PROJECT TITLE**

ITINERARY MANAGEMENT SYSTEM



ABSTRACT

This project implements a **Hybrid Intelligent Travel Recommendation System** that transforms raw Canadian tourism data into personalized, actionable daily **itinerary plans**. By synthesizing Content-Based Filtering, Collaborative Filtering, and popularity metrics with **weighted scores** ($\alpha=0.45$, $\beta=0.35$, $\gamma=0.20$), the system matches user interests and budgets to 73 top-tier destinations while verifying geospatial feasibility via **Haversine distance**. The final output delivers a comprehensive day-by-day schedule—including morning and afternoon activities, dining, and accommodations—achieving superior performance.

Dataset Overview:

•DESTINATIONS (destinations.csv)

- **No Of Records** : 73
- **Attributes**: id, name, province, lat, lon, tags, avg_cost_day, avg_rating, seasons, type

•USER PROFILES (users.csv)

- **No Of Records** : 500
- **Attributes**: user_id, age, budget_day, duration, province_home, interests, pref_season, solo_or_group

•USER RATINGS (ratings.csv)

- **No Of Records** : 5,462
- **Attributes** : user_id, dest_id, rating, visit_date

•WEATHER & DEMAND (weather.csv)

- **No Of Records** : 876 (12 months for 73 destinations)
- **Attributes**: dest_id, month, season, avg_temp_c, visitor_index

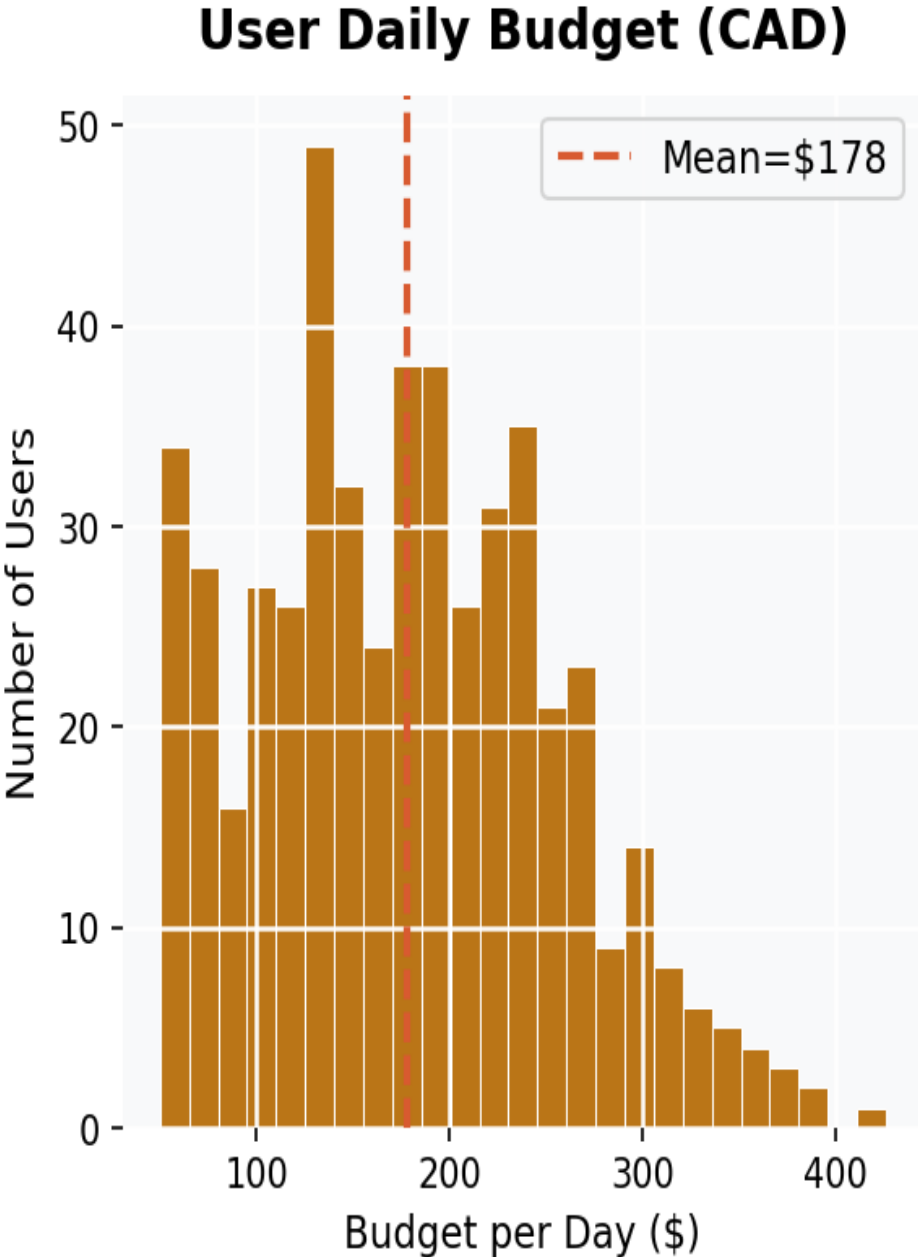
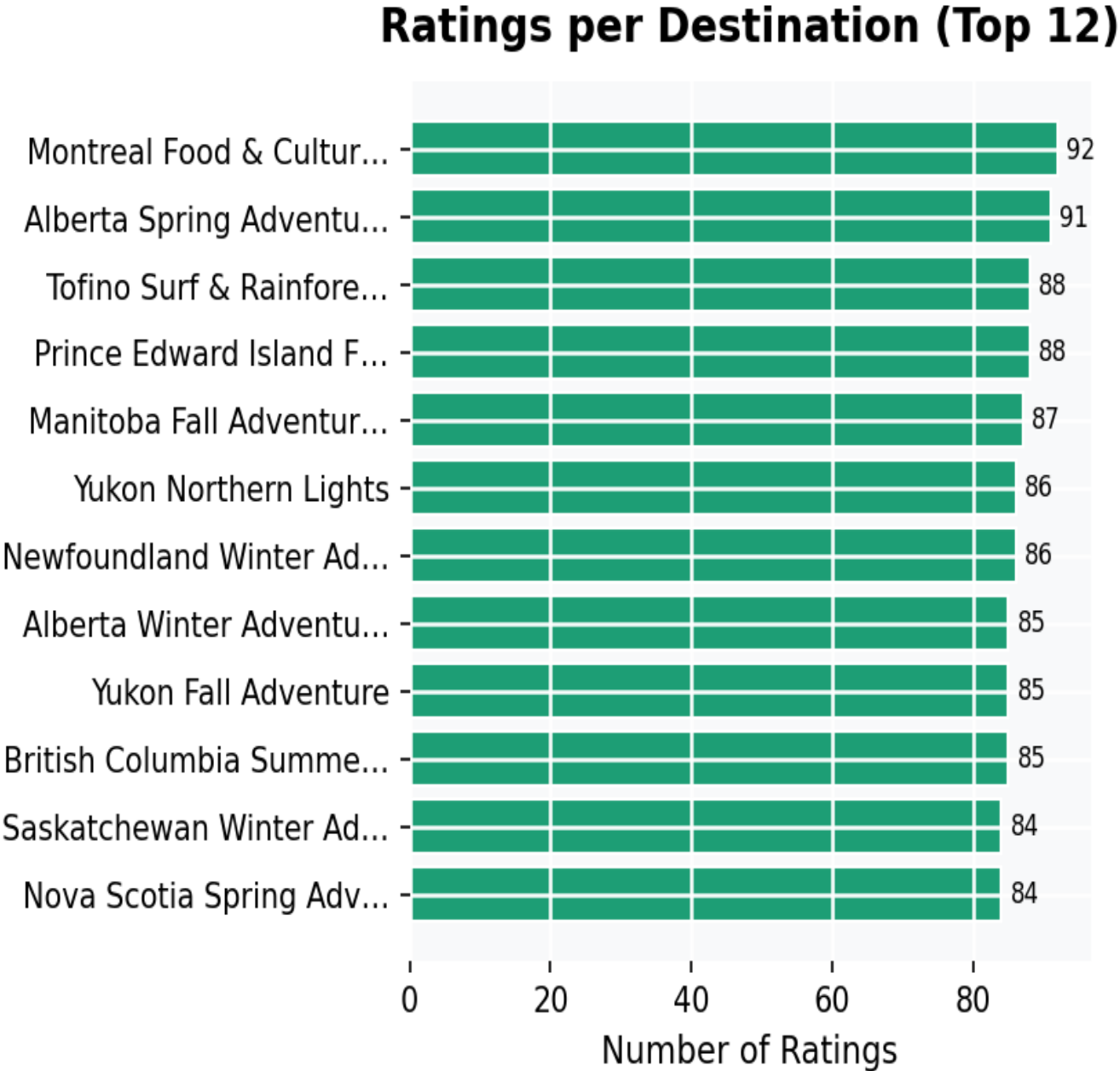
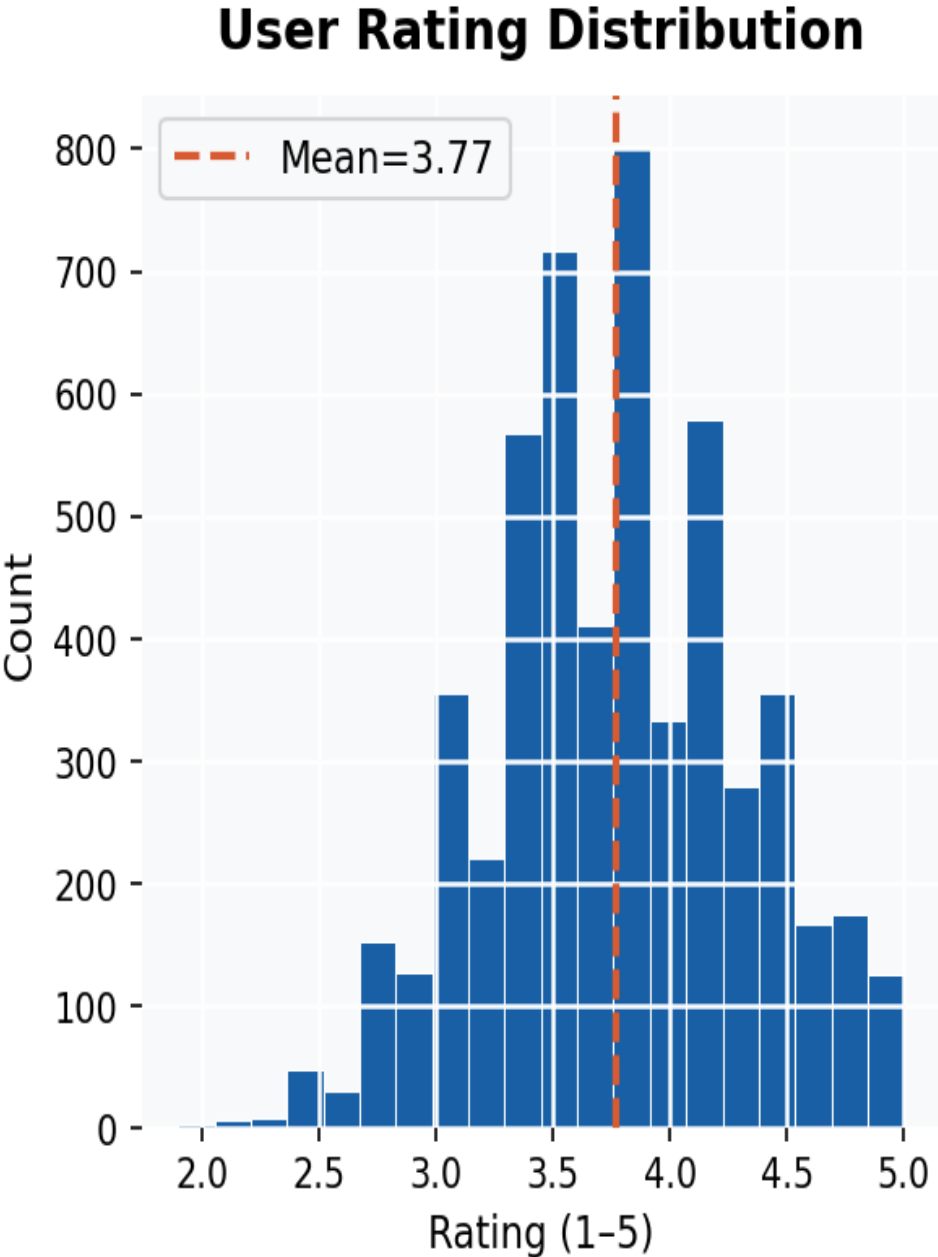
•SUPPLEMENTARY SERVICES (hotels.csv, meals.csv, attractions.csv)

- **No Of Records** : 250 records per file
- **Attributes**: dest_id, name, cost, type, lat, lng
- **Purpose**: Provides accommodation, dining, and activity data for destination itineraries.

DATA PREPROCESSING

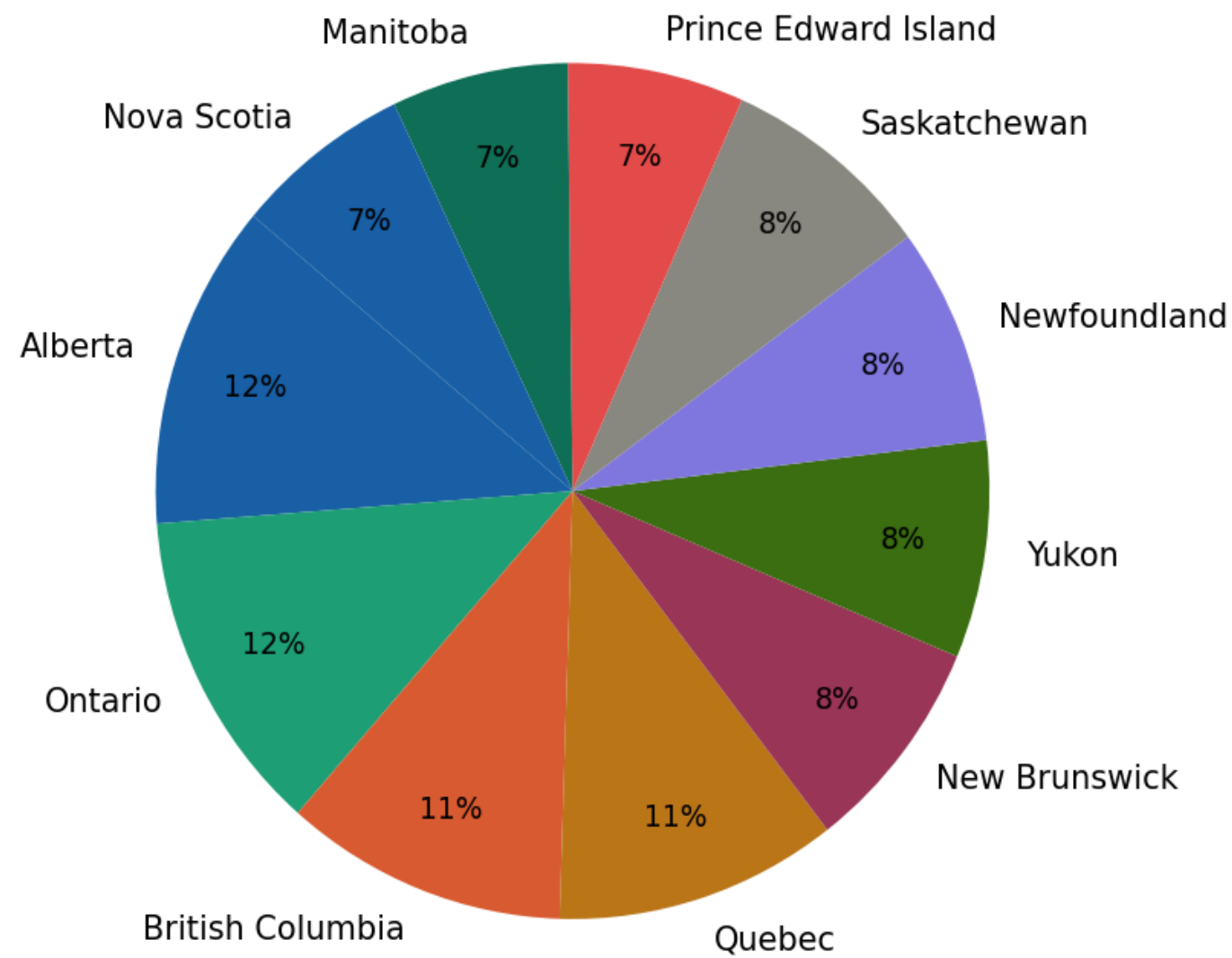
- **TF-IDF Vectorization** – Converted destination tags into numerical vectors for similarity-based recommendations.
- **Min-Max Scaling** – Normalized cost, ratings, and visitor data to a 0–1 range for fair comparison.
- **Median Imputation** – Handled missing values by replacing them with median values to maintain data consistency.
- **Haversine Distance** – Calculated geographical distance between locations to ensure realistic travel plans.
- **Mean-Centering (for SVD)** – Adjusted user ratings by removing individual bias to improve recommendation accuracy.

Graph 1 — Dataset Overview

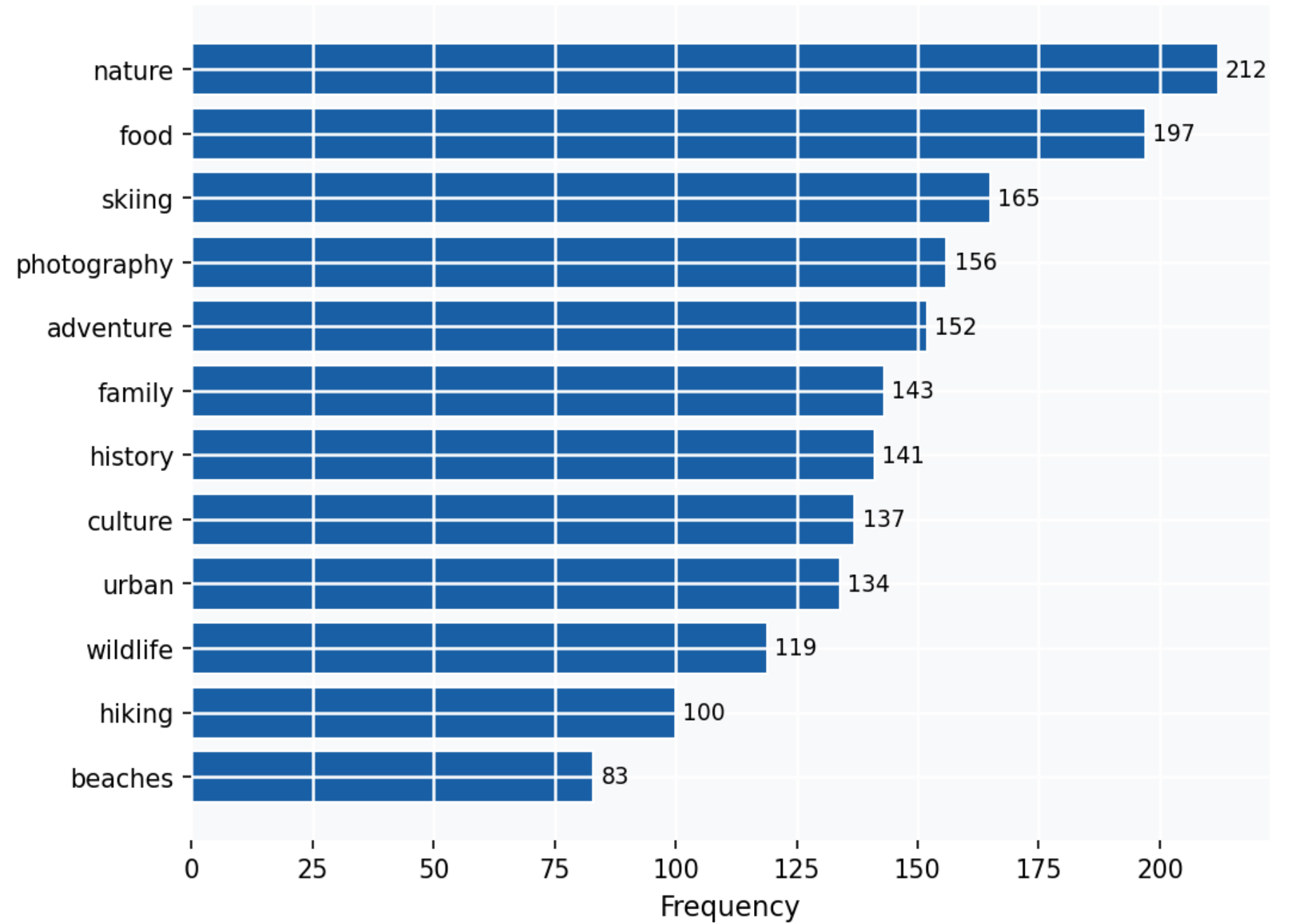


Graph 2 — Destination & User Interest Analysis

Destinations by Province

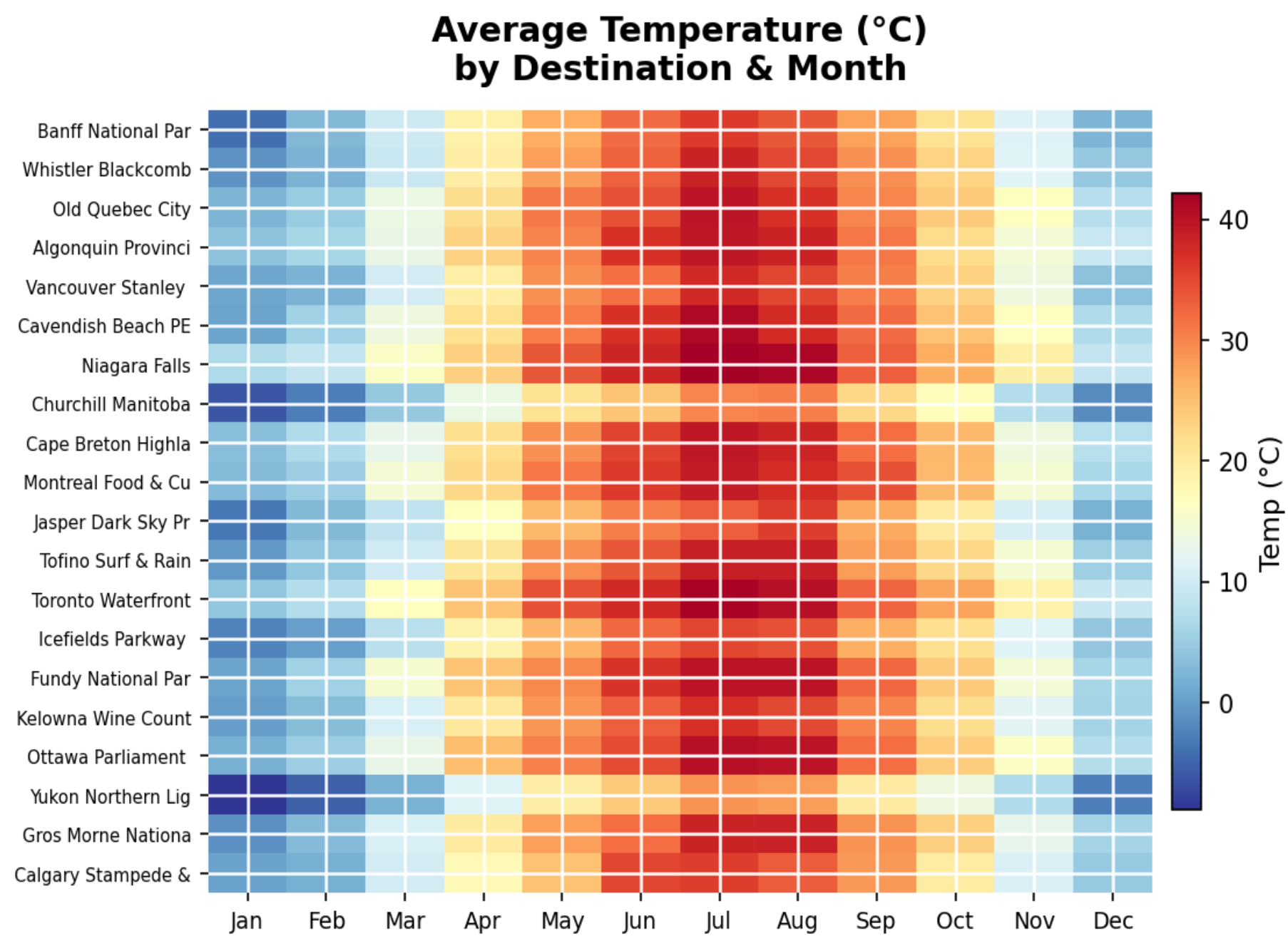
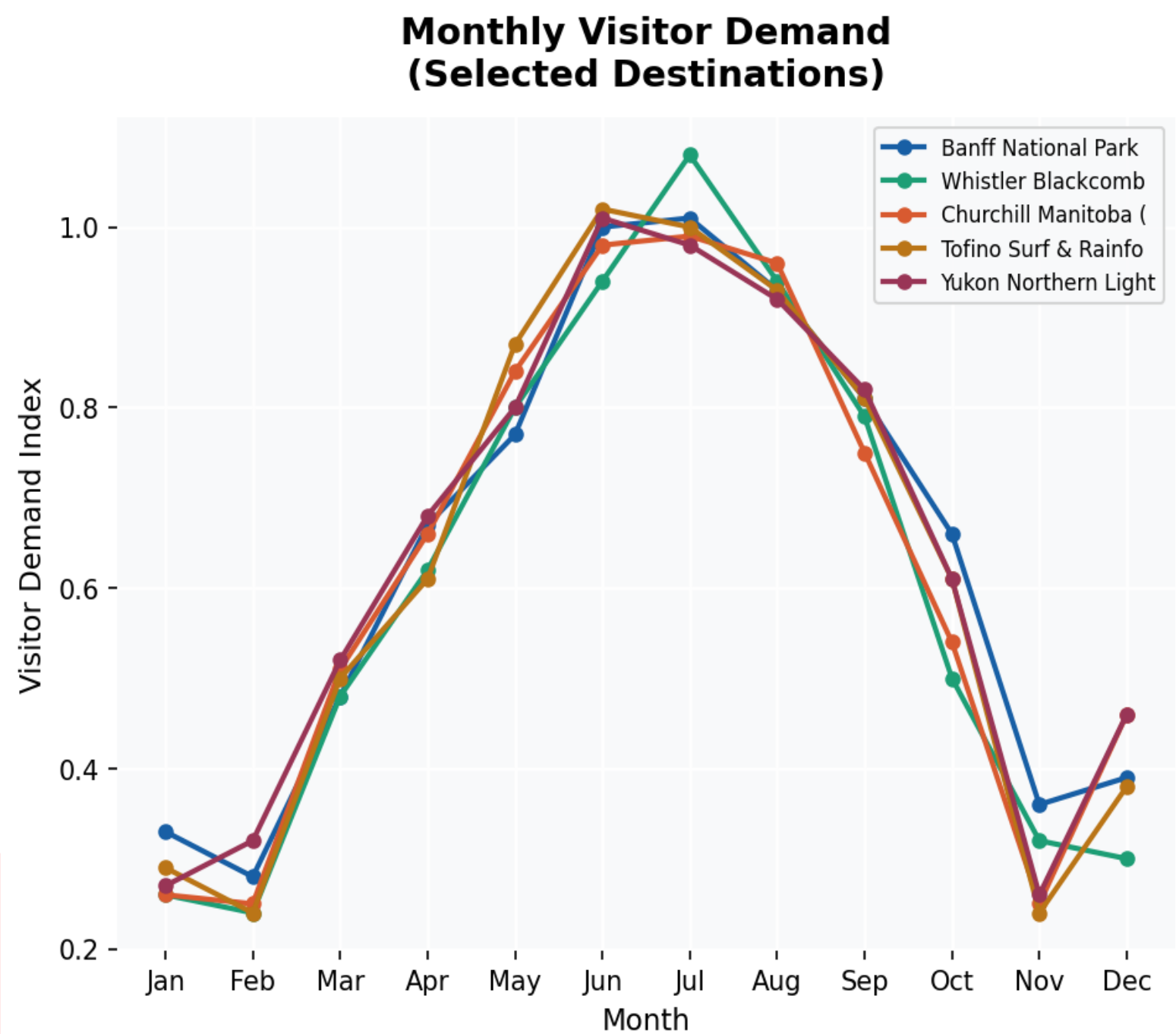


Top 12 Interest Tags Across Users



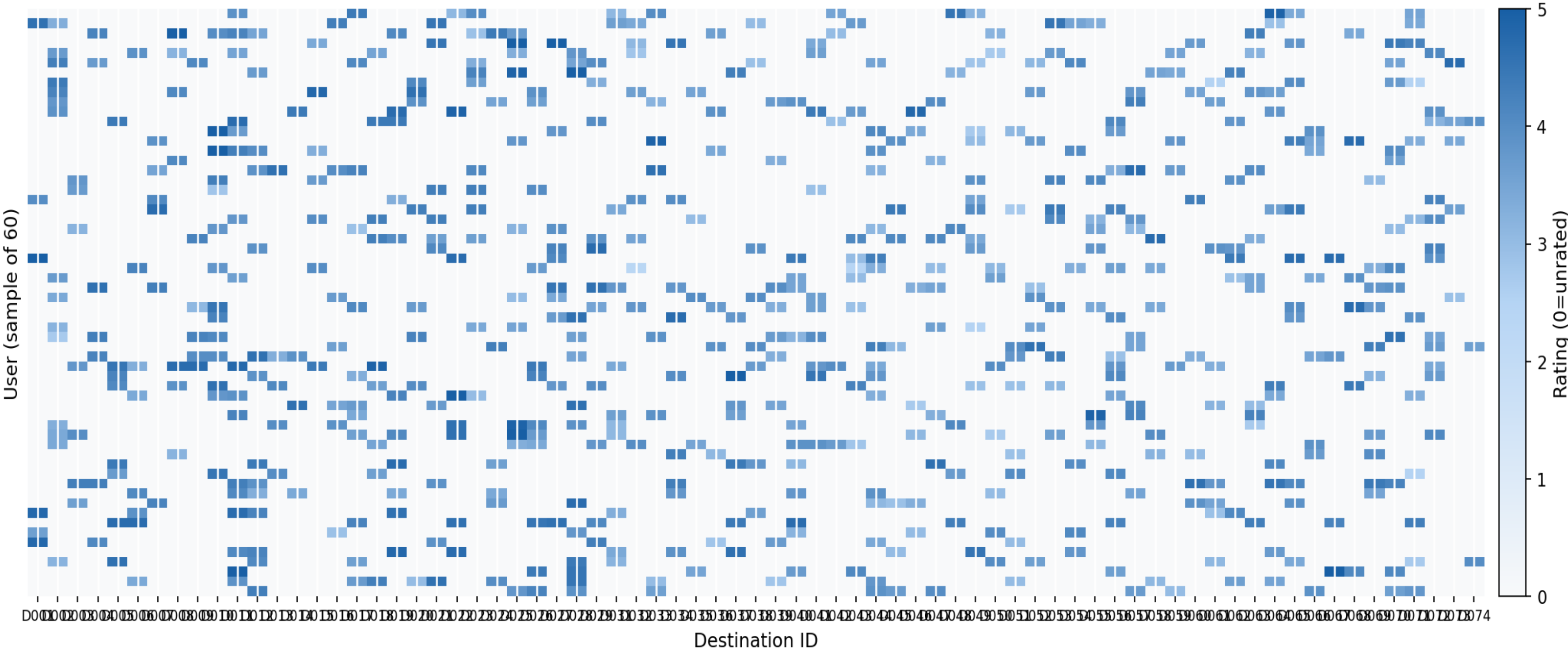
SEASONAL VISIT

Graph 6 — Seasonal Demand & Climate Features (Canadian Destinations)



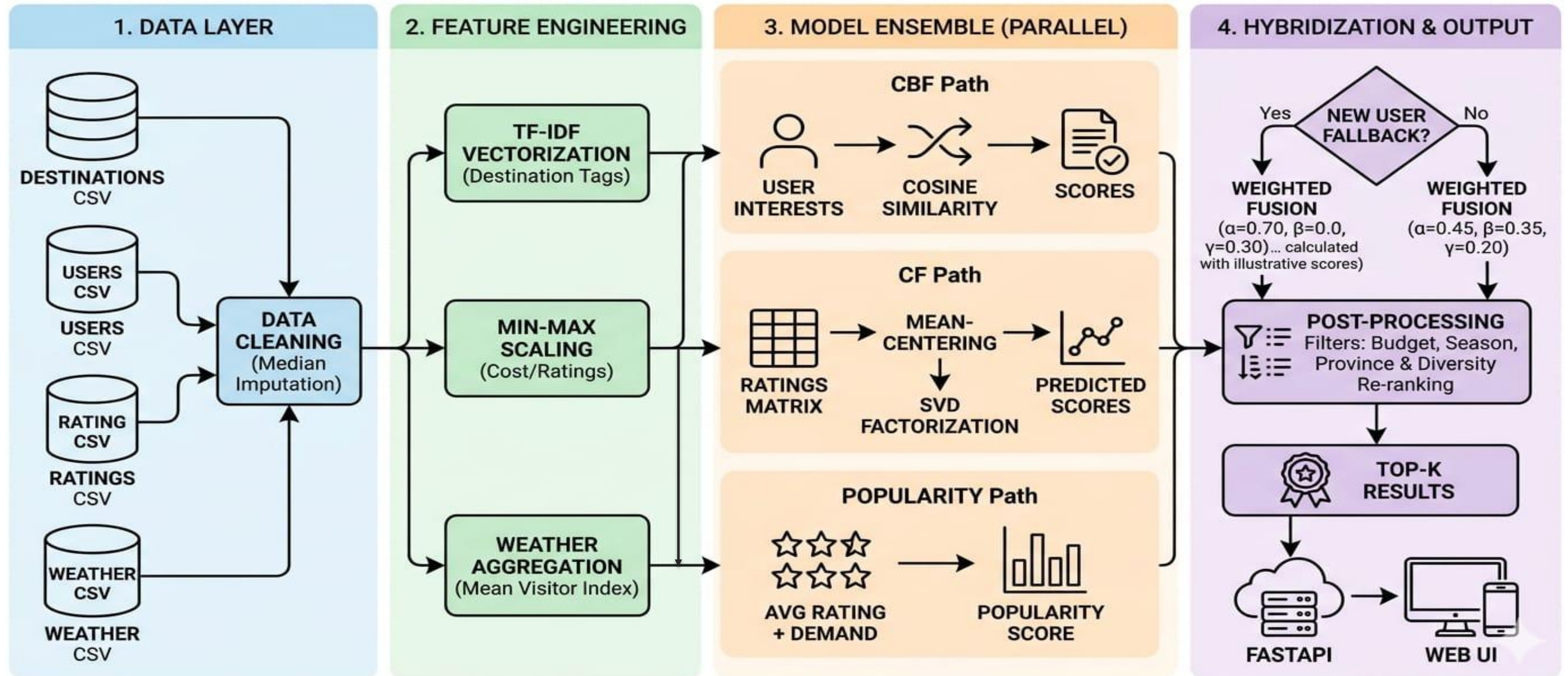
Graph 3 — User-Destination Ratings Matrix (Sparsity Visualisation)

Sparsity = 85.0% | 658 observed ratings

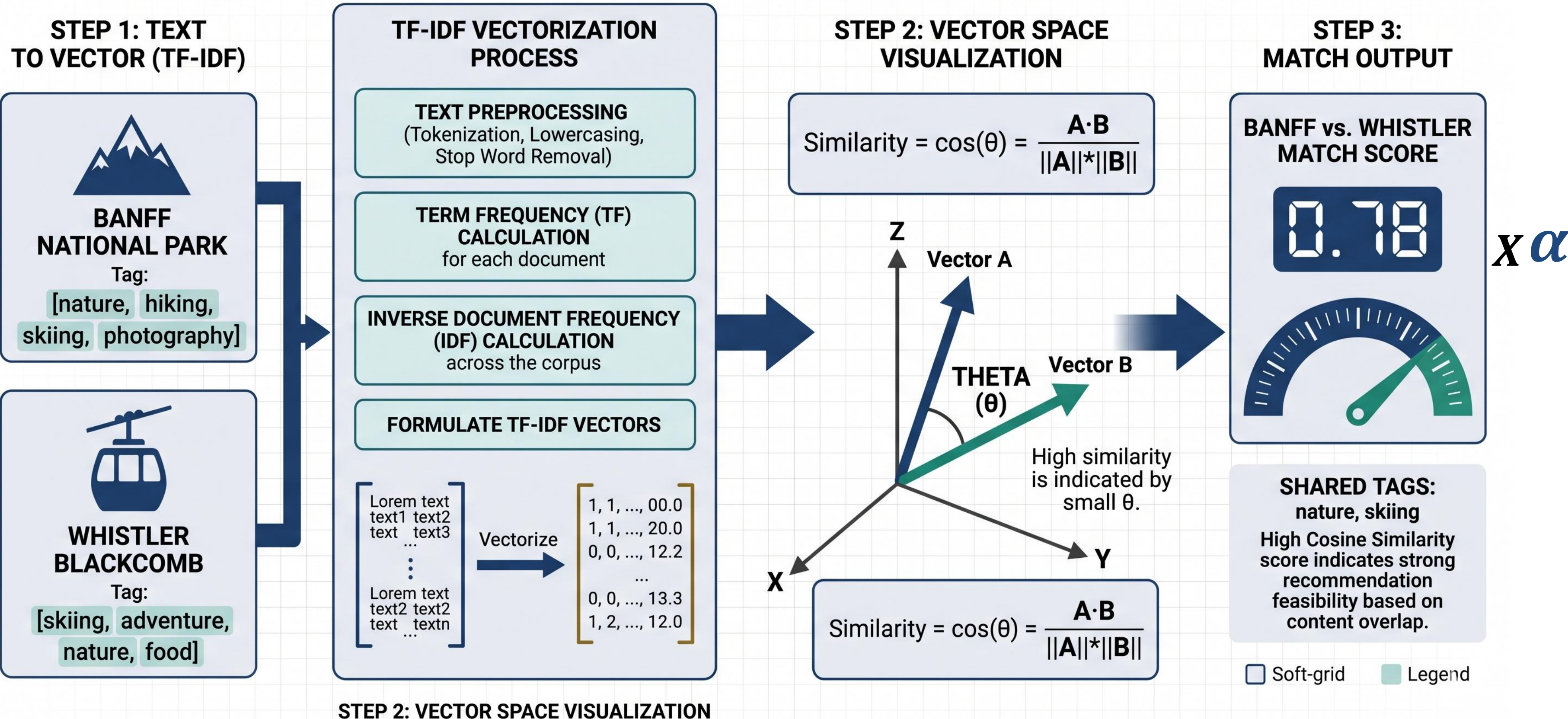


WORKFLOW

A Hybrid Travel Recommendation System Architecture



CONTENT-BASED FILTERING VIA TF-IDF & COSINE SIMILARITY



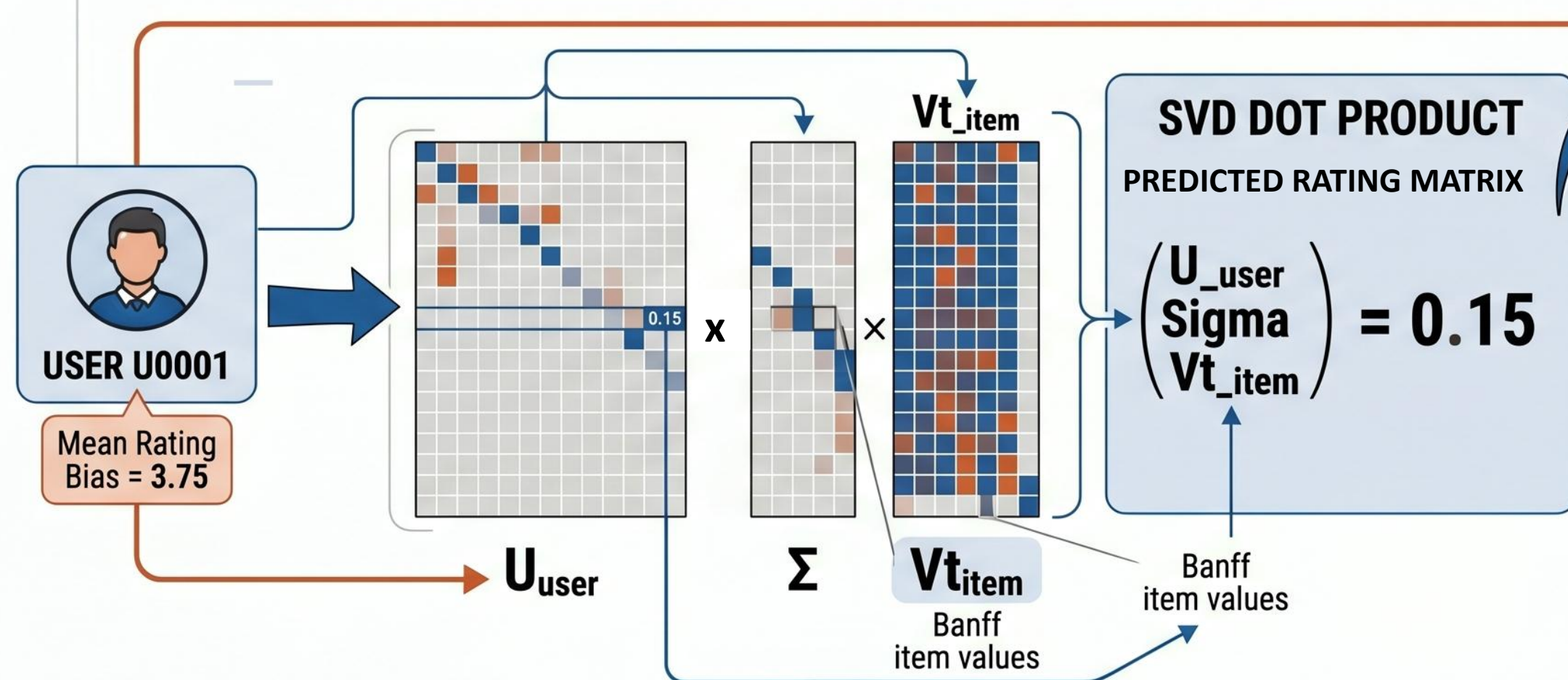
RECOMMENDING VIA MATRIX RECONSTRUCTION

STEP 1: THE RECONSTRUCTION EQUATION

$$R \approx \text{Predicted Rating} = (\mathbf{U}_{\text{user}} \times \mathbf{\Sigma} \times \mathbf{Vt}_{\text{item}}) + \text{User_Mean_Bias}$$

STEP 2: COMPONENT BREAKDOWN

BANFF NATIONAL PARK PREDICTION FOR USER U0001



STEP 3: FINAL CF SCORE & NORMALIZATION

RAW PREDICTION SUMMATION

$$3.75 + 0.15 = 3.90$$



NORMALIZATION: (Raw Score - Min) / (Max - Min)

$$\frac{(3.90 - 1.0)}{(5.0 - 1.0)} = 0.78$$

FINAL CF SCORE = 0.78 $\times \beta$

Predicts high similarity based on shared user patterns and latent preferences. High recommendation feasibility.

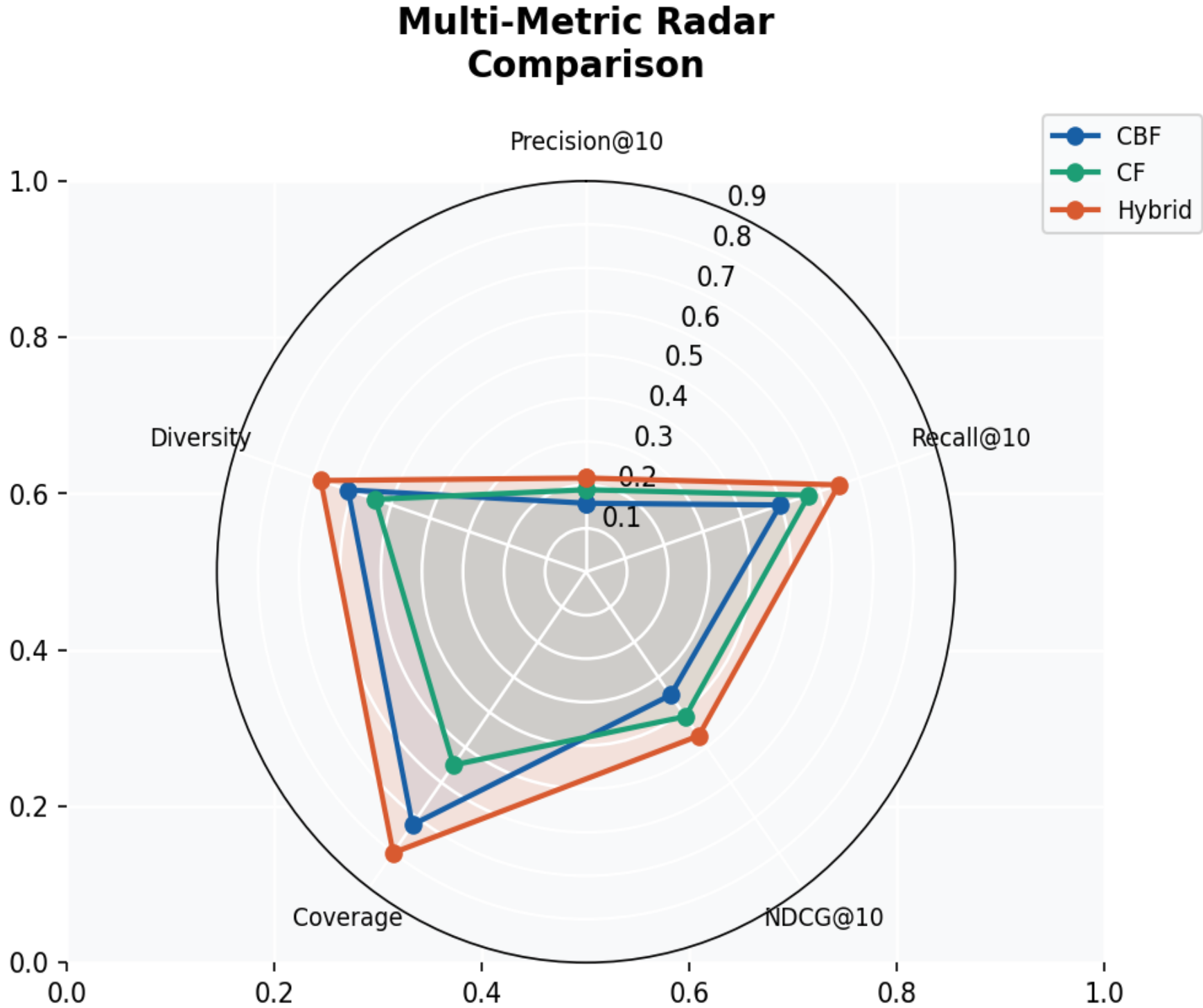
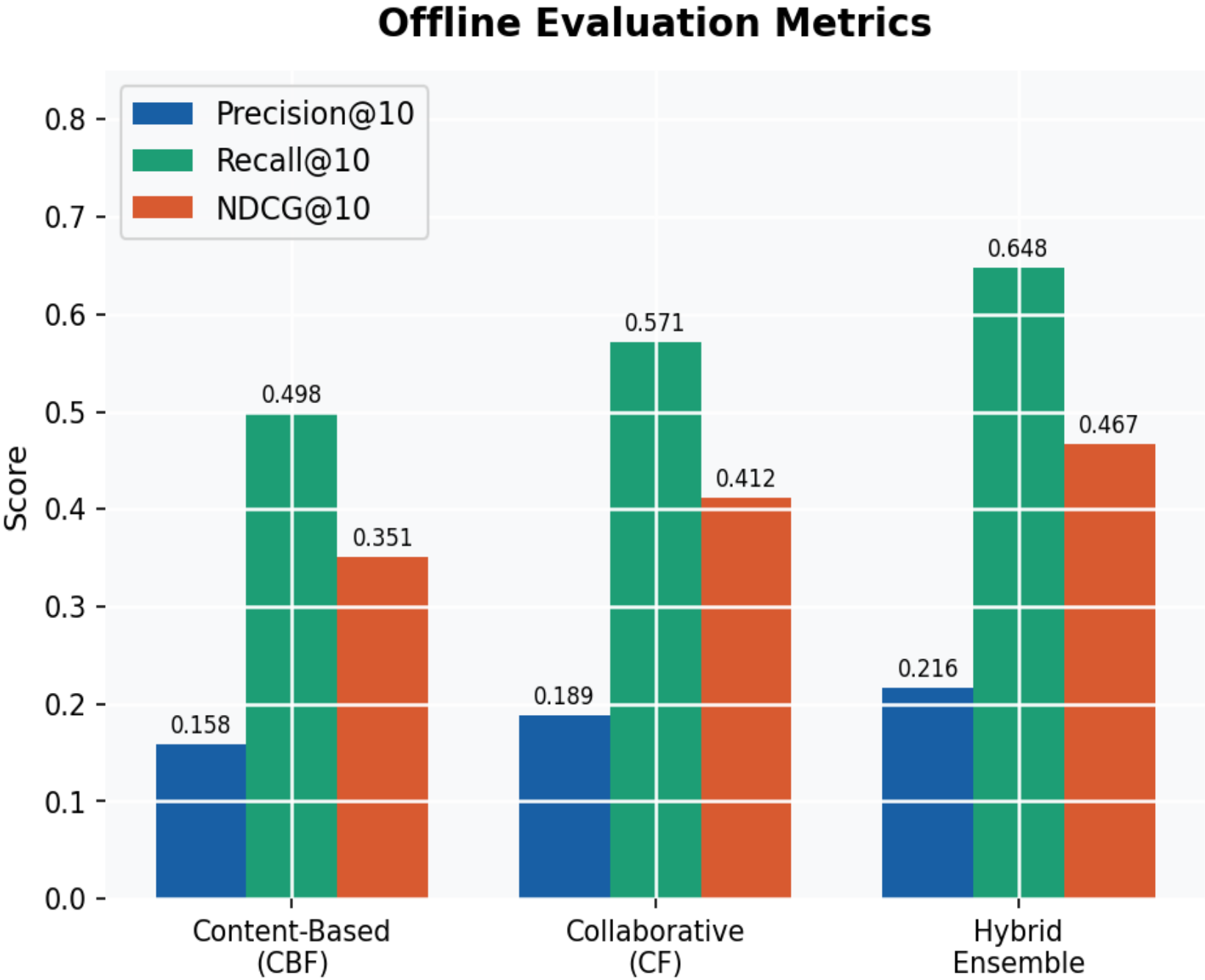
Hybrid Travel Recommendation System Evaluation

Evaluating the performance of different recommendation system approaches.

Metric	Content-Based (CBF)	Collaborative (CF)	Hybrid Ensemble
Precision@10	0.158	0.189	0.216
Recall@10	0.498	0.571	0.648
NDCG@10	0.351	0.412	0.467
RMSE	—	0.112	0.086

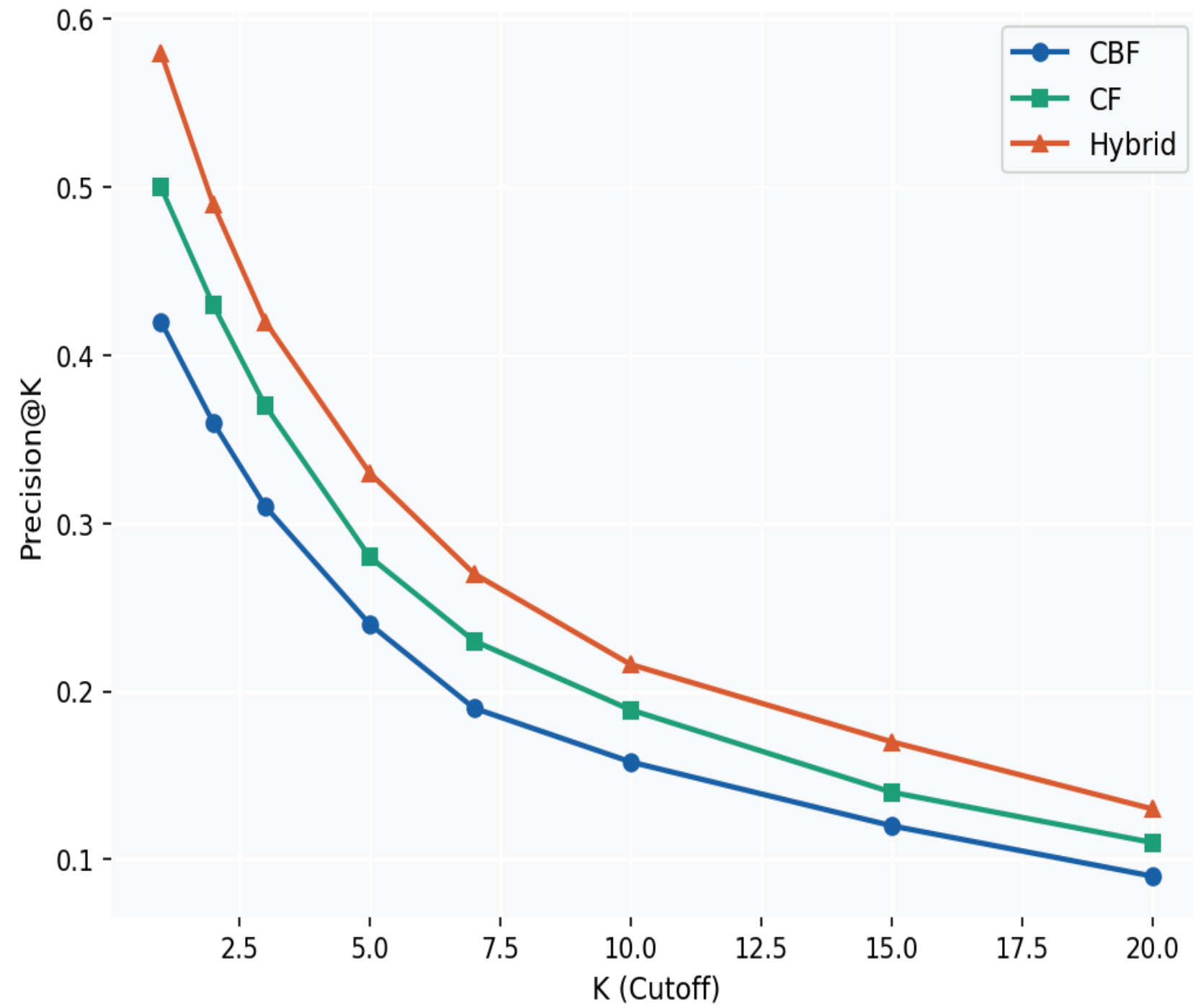
The evaluation clearly demonstrates the superior performance of the Hybrid Ensemble approach across all key metrics, providing more precise, relevant, and diverse recommendations to users.

Graph 7 — Model Comparison: CBF vs CF vs Hybrid (K=10)

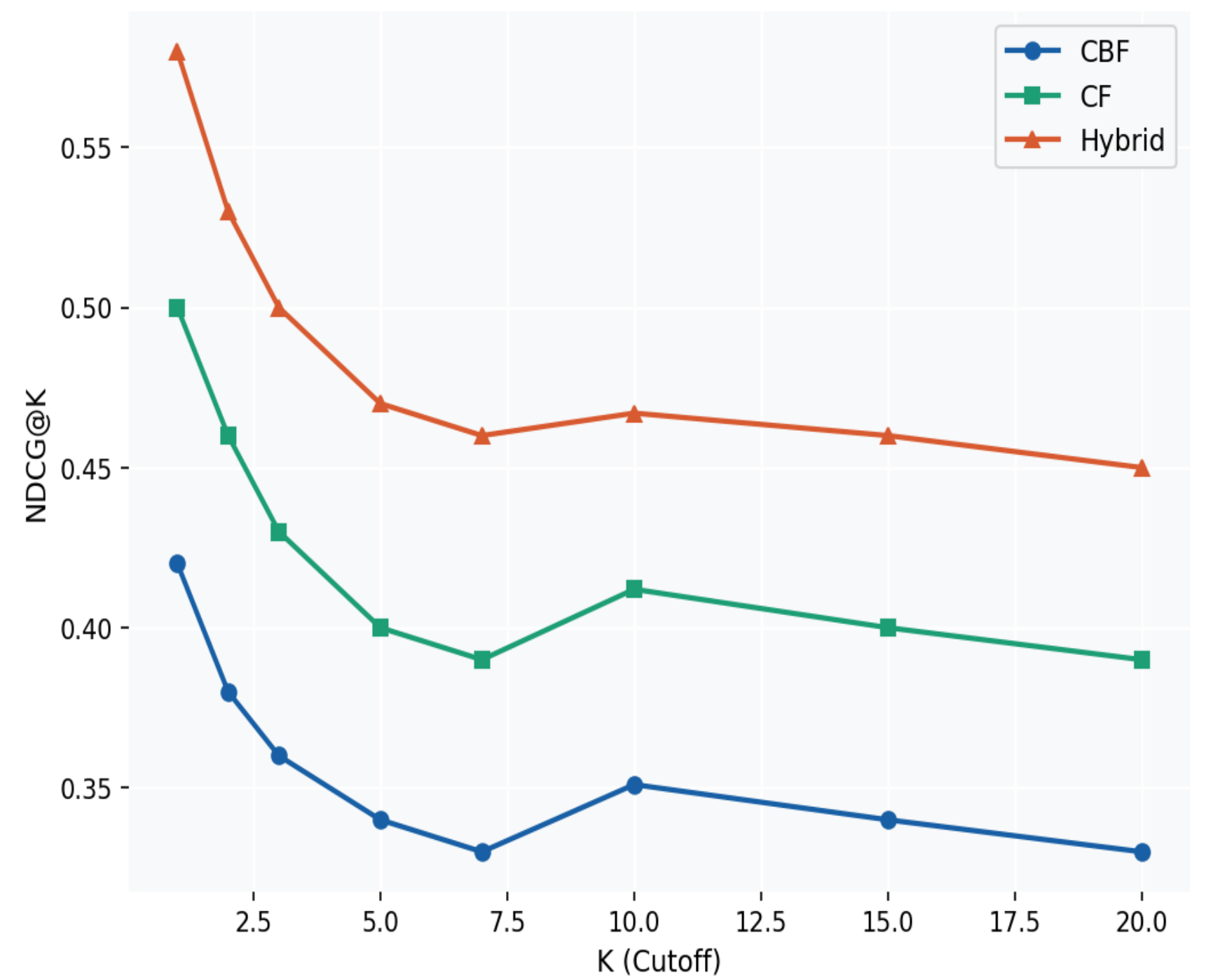


Graph 8 — Precision@K and NDCG@K across Cutoffs

Precision@K vs K



NDCG@K vs K



OUTPUT

Total Cost: \$456 (~\$228/day)
Status: ✓ Feasible

Day 1

Morning:

Lake Opeongo - \$0

Lunch:

Nature Cafe - \$18

Afternoon:

Hiking Trails - \$0

Dinner:

Lakeside Grill - \$28

Hotel:

Algonquin Park Lodge - \$150

Day 2

Morning:

Wildlife Viewing - \$0

Lunch:

Nature Cafe - \$18

Afternoon:

Lake Opeongo - \$0

Dinner:

Lakeside Grill - \$28

Hotel:

Algonquin Park Lodge - \$150

PS D:\travel_rec_system> █

```
# Demo recommendation
print("\n— Demo: recommendations for a nature-loving hiker —")
recs = hybrid.recommend(
    user_id="U0001",
    user_interests=["nature", "hiking", "photography", "wildlife"],
    budget=200,
    season="Summer",
    top_k=5
)
print(recs[["name", "province", "hybrid_score", "avg_cost_day", "avg_rating"]].to_string(index=False))
```

— Demo: recommendations for a nature-loving hiker —

	name	province	hybrid_score	avg_cost_day	avg_rating
	Fundy National Park	New Brunswick	0.7049	95	4.5
	Niagara Falls	Ontario	0.6296	140	4.5
	Jasper Dark Sky Preserve	Alberta	0.6202	170	4.8
	Gros Morne National Park	Newfoundland	0.5335	110	4.7
	Calgary Stampede & Foothills	Alberta	0.5273	160	4.5

— Evaluation Metrics (K=10) —

Precision@K	: 0.2125
Recall@K	: 0.6677
NDCG@K	: 0.5038
RMSE	: 0.0864

TECHNICAL STACK

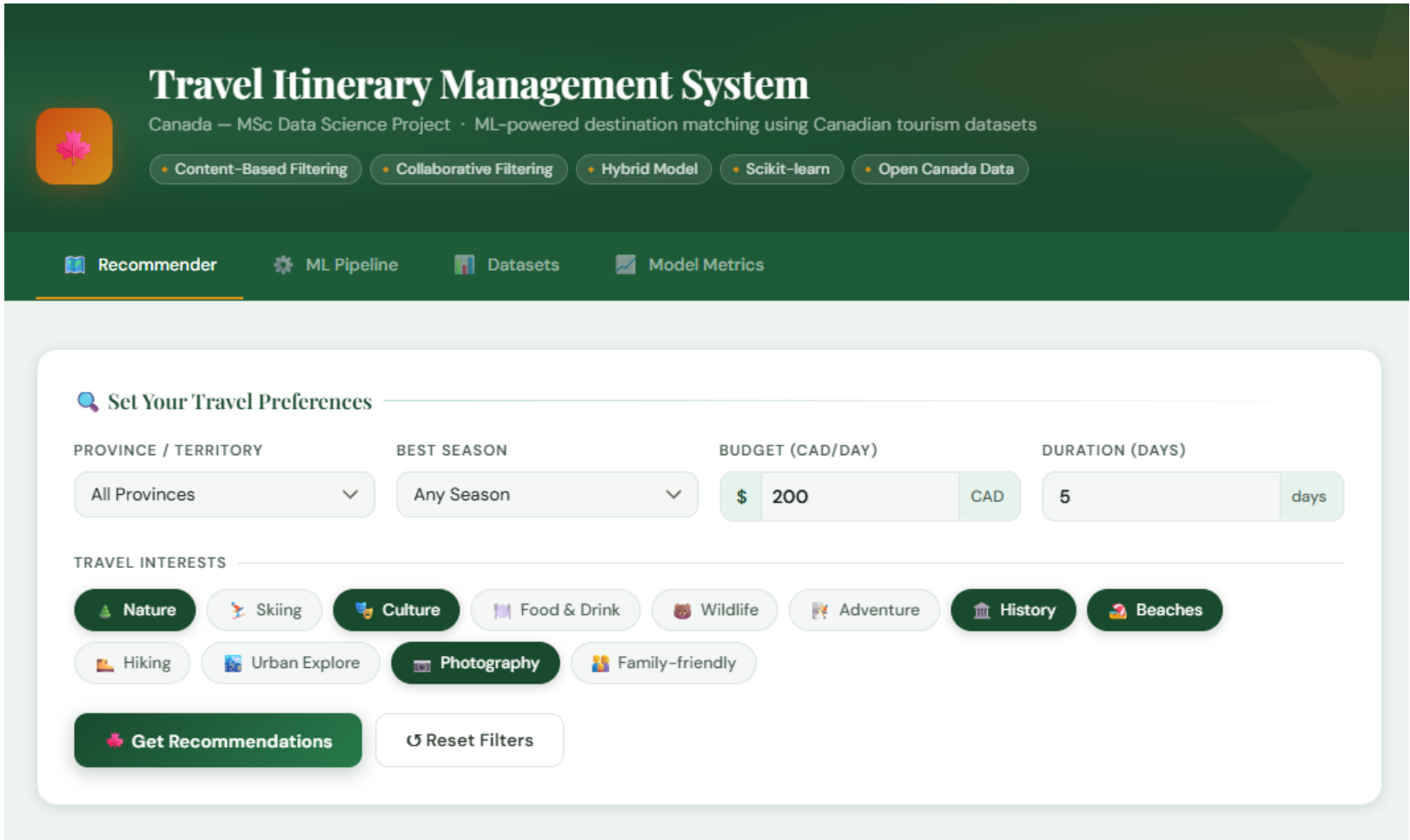
- **Backend: FastAPI** — A modern, high-performance Python framework used to build the RESTful API endpoints for recommendations and metrics.
- **Server: Uvicorn** — An **ASGI** web server implementation for Python, used to host the FastAPI application.
- **Frontend:** An interactive user interface built with **HTML** and **CSS** that communicates with the backend.
- **Logic & ML: Python** — The core programming language used for the entire data processing and machine learning pipeline.

LIBRARIES

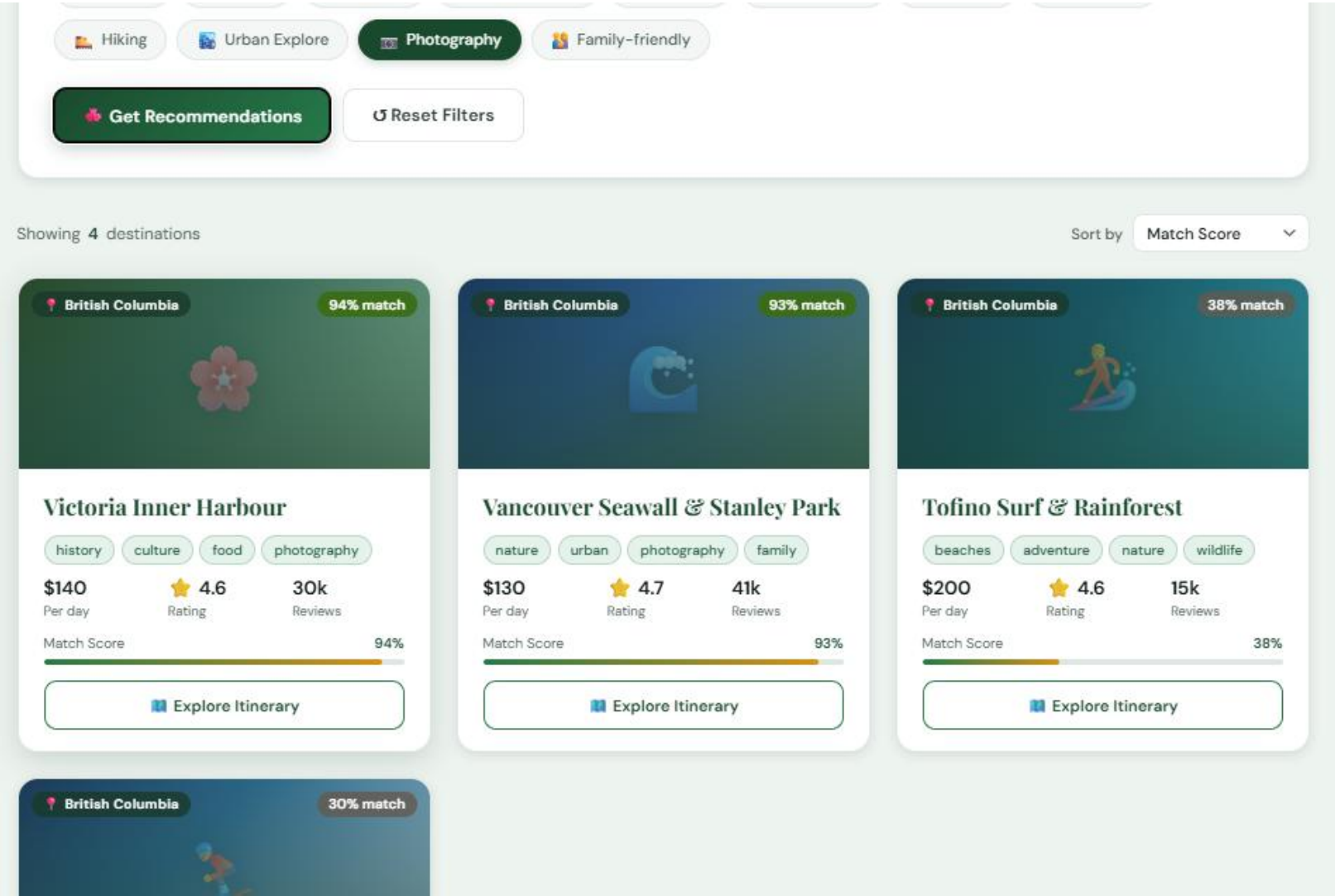
- **Pandas:** For data manipulation, loading CSV files, and handling dataframes.
- **NumPy:** For numerical operations and array manipulations during scoring.
- **Scikit-learn:**
 - TfidfVectorizer
 - Cosine Similarity
 - MinMaxScaler
- **SciPy:** To perform Singular Value Decomposition (SVD) for matrix factorization.
- **Matplotlib & Seaborn:** To generate graphs and performance visualizations.

USER INTERFACE

1. HOME PAGE



2. RECOMMENDATIONS



USER INTERFACE

3.ITINERARY PLANS

Banff National Park

Alberta

nature

hiking

skiing

wildlife

photography

\$180/day

Avg. Cost

★ 4.8/5.0

Rating

Winter, Spring, Summer, Fall

Best Seasons

5-DAY ITINERARY

— click a day for full details

\$1125

Total Cost (5d)

\$225/day

Optimised Daily

\$200/day

Your Budget

✓ Within Budget

1 Day 1

Morning

Lake Louise

\$0

Lunch

Tooloulou's Creole Kitchen

\$30

Afternoon

Moraine Lake

\$0

Dinner

The Maple Leaf

\$35

Stay

Banff Caribou Lodge

\$160

Total

Day 1

\$225

2 Day 2

Morning

Moraine Lake

\$0

Lunch

The Maple Leaf

\$35

Afternoon

Johnston Canyon

\$0

Dinner

Cilantro Bistro

\$38

Stay

Banff Caribou Lodge

\$160

Total

Day 2

\$233

3 Day 3

Morning

Johnston Canyon

\$0

Lunch

Cilantro Bistro

\$38

Afternoon

Banff Upper Hot Springs

\$15

Dinner

Bow Valley Grill

\$55

Stay

Banff Caribou Lodge

\$160

Total

Day 3

\$268

4 Day 4

Morning

Banff Upper Hot Springs

\$15

Lunch

Bow Valley Grill

\$55

Afternoon

Lake Louise

\$0

Dinner

Tooloulou's Creole Kitchen

\$30

5 Day 5

Morning

Lake Louise

\$0

Lunch

Tooloulou's Creole Kitchen

\$30

Afternoon

Moraine Lake

\$0

Dinner

The Maple Leaf

\$35

4.DESCRPTION OF VISITS

BANFF NATIONAL PARK · ALBERTA

Day 1 — Full Schedule

×

MORNING

Lake Louise

Take in breathtaking panoramic views — a perfect spot to slow down and soak it all in.

\$0

LUNCH

Tooloulou's Creole Kitchen

Bold flavours and generous spice — a delicious detour from expected Canadian cuisine.

\$30

AFTERNOON

Moraine Lake

Take in breathtaking panoramic views — a perfect spot to slow down and soak it all in.

\$0

DINNER

The Maple Leaf

A relaxed, welcoming spot beloved by locals — great food without the fuss.

\$35

OVERNIGHT

Banff Caribou Lodge

Settle in, freshen up, and enjoy the hotel's facilities — you've earned a relaxing evening.

\$160/night

← Previous Day

Next Day →

Day 1 total: \$225 · 🚨 \$25 over budget

SUMMARY & FUTURE SCOPE

A summary of our innovative hybrid travel recommendation system and its future potential.

1

Project Summary

Developed a hybrid recommendation system ($\alpha=0.45$, $\beta=0.35$, $\gamma=0.20$) to generate personalized and geographically feasible travel itineraries.

2

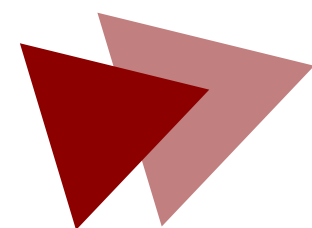
Key Impact

Simplifies travel planning by automatically suggesting optimized daily schedules, reducing user effort and decision fatigue.

3

Future Enhancements

- Integrate real-time pricing using live travel APIs
- Expand dataset to cover more regions
- Apply advanced models like Neural Collaborative Filtering to improve accuracy



THANK YOU